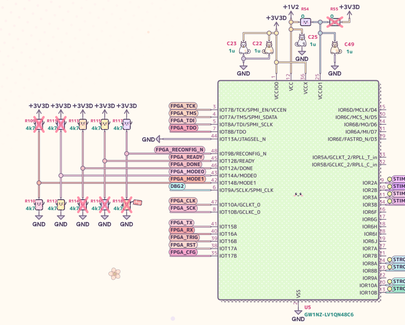
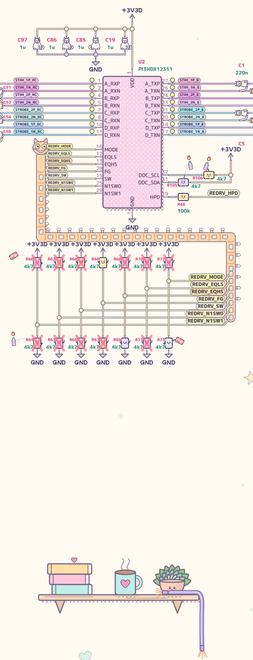
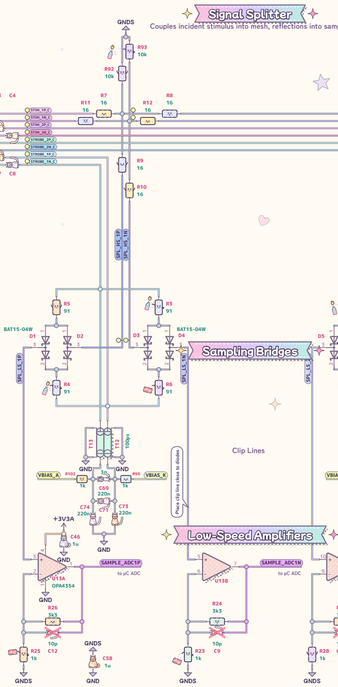
FPGA
Stimulus & Sampling Pulse Timing Generation



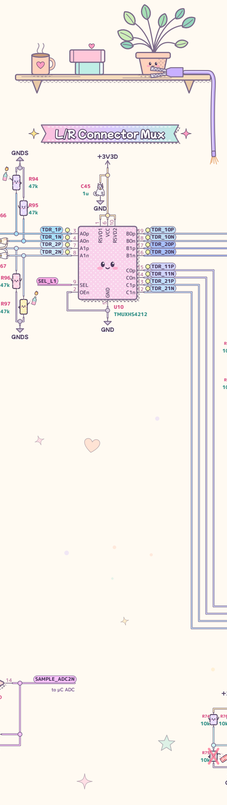
Pulse Amplifier



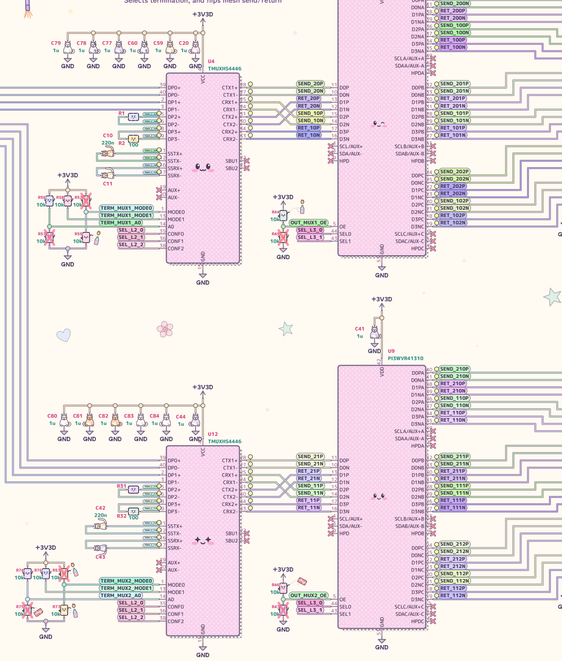
Signal Splitter
Couples incident stimulus into mesh, reflections into sampler



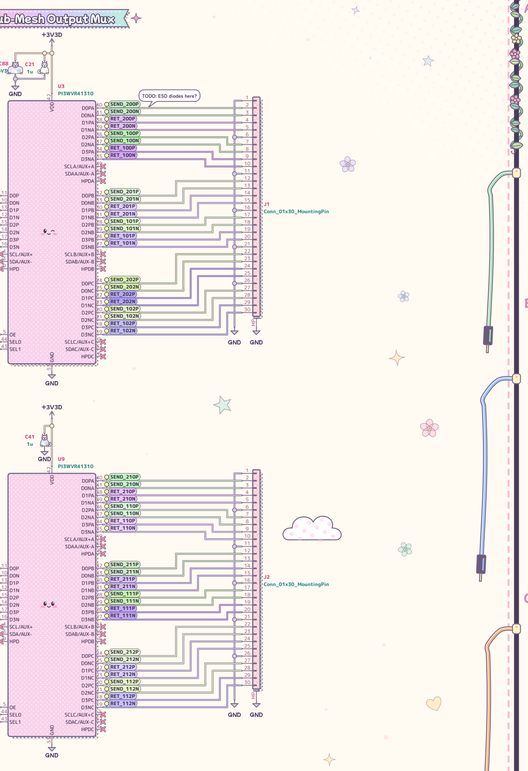
L/R Connector Mux



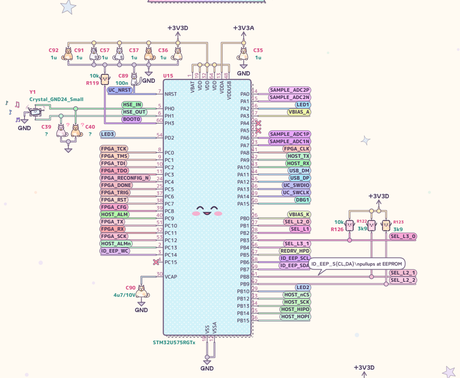
Termination Mux & Crossbar
Selects termination, and flips mesh send/return



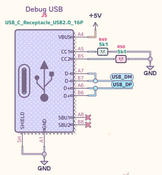
Sub-Mesh Output Mux

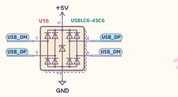


Microcontroller

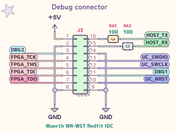


Debug USB

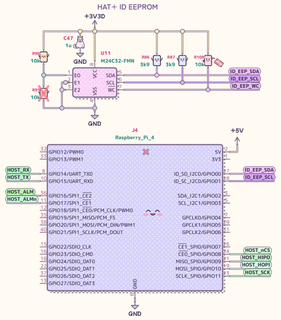




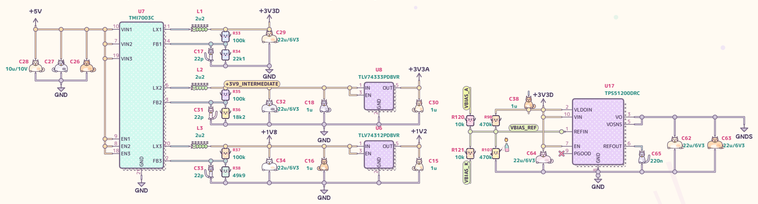
Debug connector



Pi HAT+ Connector



Power Supplies
5V input, outputs: 3V3D (digital), 3V3A (analog, low noise), 1V8 (digital), 1V2 (digital)



H2 M2.5 H3 M2.5 H4 M2.5 H5 M2.5

♡ cast of characters ♡

10pF 1nF 100nF 1µF 4µ7F 22µF

capacitors
more capacitance, more chonk
(the wires go in one end and out the other)

resistor
grumpy, throws bricks
(and the odd molotov)

inductor
a very hungry caterpillar

crystal
dances at its own frequency ♪

bus
a very long cat
carries lots of signals

キラキラ kira kira ♡
ドキドキ doki doki ♡
ふわふわ fuwa fuwa ♡

let's sense some pillows~! ♡

✦ センサー回路図 ✦

yanartas-pillowbox-sensors

~ the pillowbox sensor board ~

sheet ♡ 1

paper A1 · drawn with love ♪